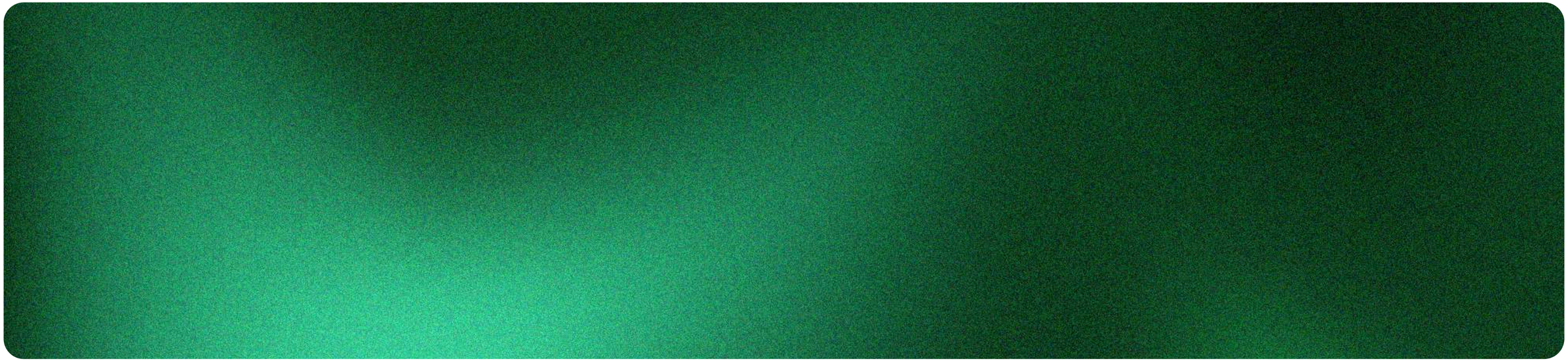
Handmade E-Commerce Sales Analysis Portfolio

Statistical EDA, Trends, & Top Markets Insights Using Python

Created By: **Shalsabila Sofiana Nur**



Project Overview

What is this Project about?

An analysis of the statistical and business aspects of handicraft e-commerce sales in 2011. The company is based in the UK.

What problem does it solve?

This project tries to test the reliability of the data and discover the pattern of the company's monthly revenue, the highest market, and the most popular products in 2011.

What kind of tool that was used?

From CSV dataset processed with Python using Pandas, Numpy, Matplotlib, and Seaborn Library.



Project Summary

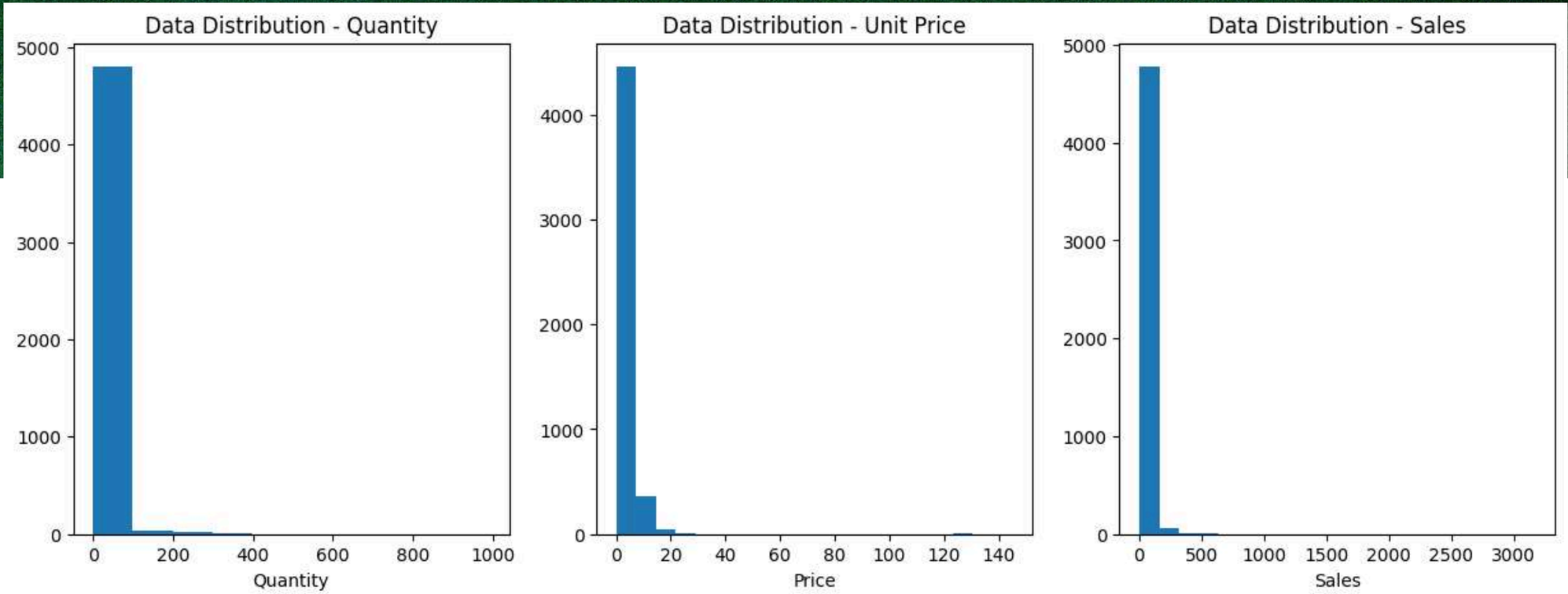
EDA

1. The metric datasets, such as quantity, unit price, and sales data are distributed heavily towards right skew
2. There are outliers in 3 data metrics and they make up less than 10%.
3. There is a strong positive correlation between Quantity and Sales.
4. There is no correlation between Quantity vs Unit Price and Unit Price vs Sales.

Business Analysis

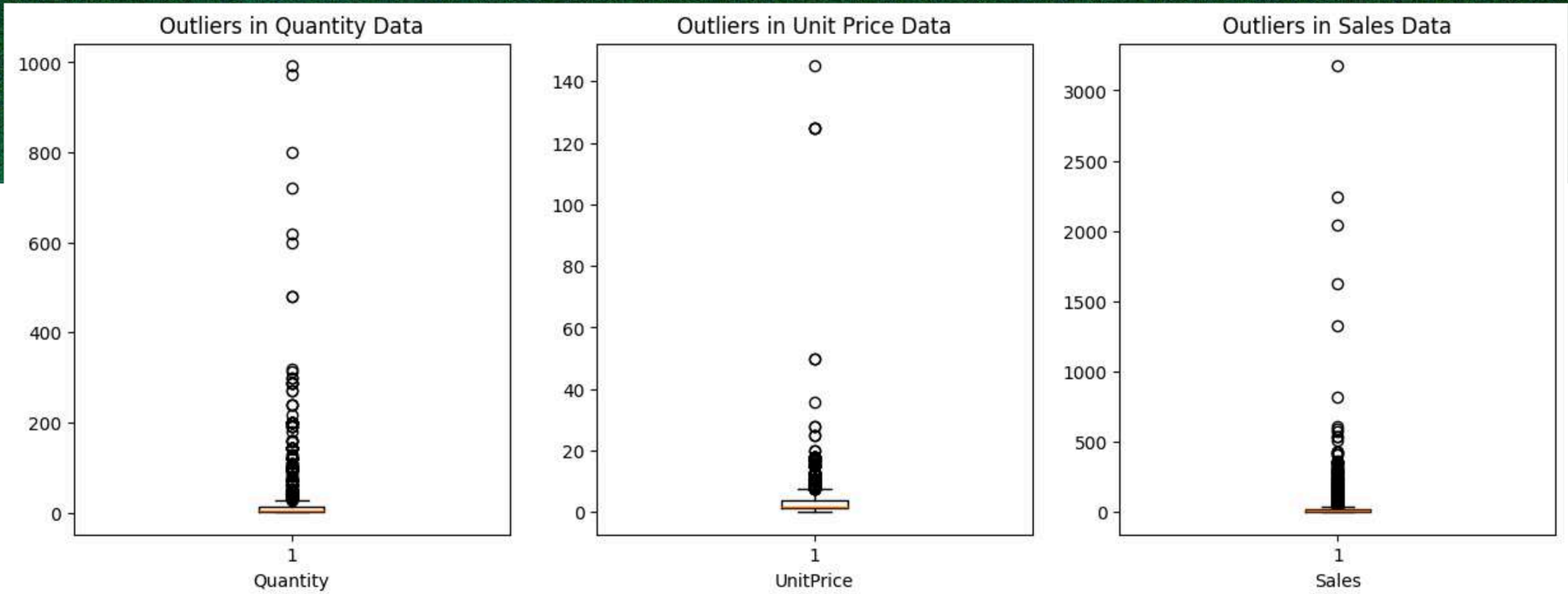
1. Generally, in 2011, the company showed very significant growth, especially on November, when sales reached its peak at nearly 18,000 USD.
2. Top 5 countries with the most sales are UK, Ireland, Germany, France, and the Netherlands, with UK as the market leader.
3. There were no sales during July and December in Netherlands.
4. White Hanging Heart T-Light Holder is the best-selling product but most purchases were made by British and Irish customers.
5. Both France and Germany still keen on buying postages product.

Data Distributed towards Right Skew



The histogram shows that the data for unit price, quantity, and sales are right-skewed. This means that **the data are centered on low purchases with lower prices**, yet the frequency of the transactions are quite often.

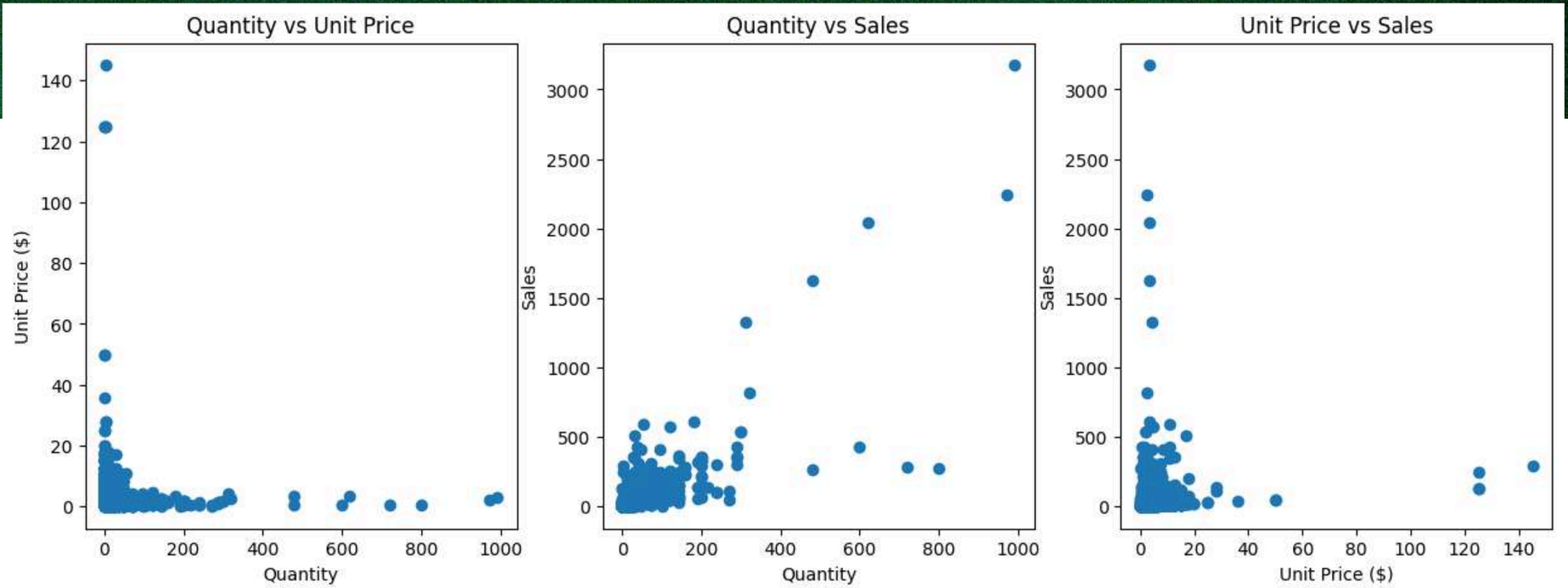
Exposing Outliers Using Boxplot



The boxplot showed outliers in unit price, quantity, and sales data.
IQR was used to **measure the data's midspread and set boundaries**. When the standard deviations surpass either lower or upper lines, meaning that there are some data points exceeding the normal boundaries.

Overall, the outliers make up **less than 10%** of the total data, which means the amounts are still tolerable and will not make the data less accurate.

Positive Correlation Only Shown Between Quantity and Sales



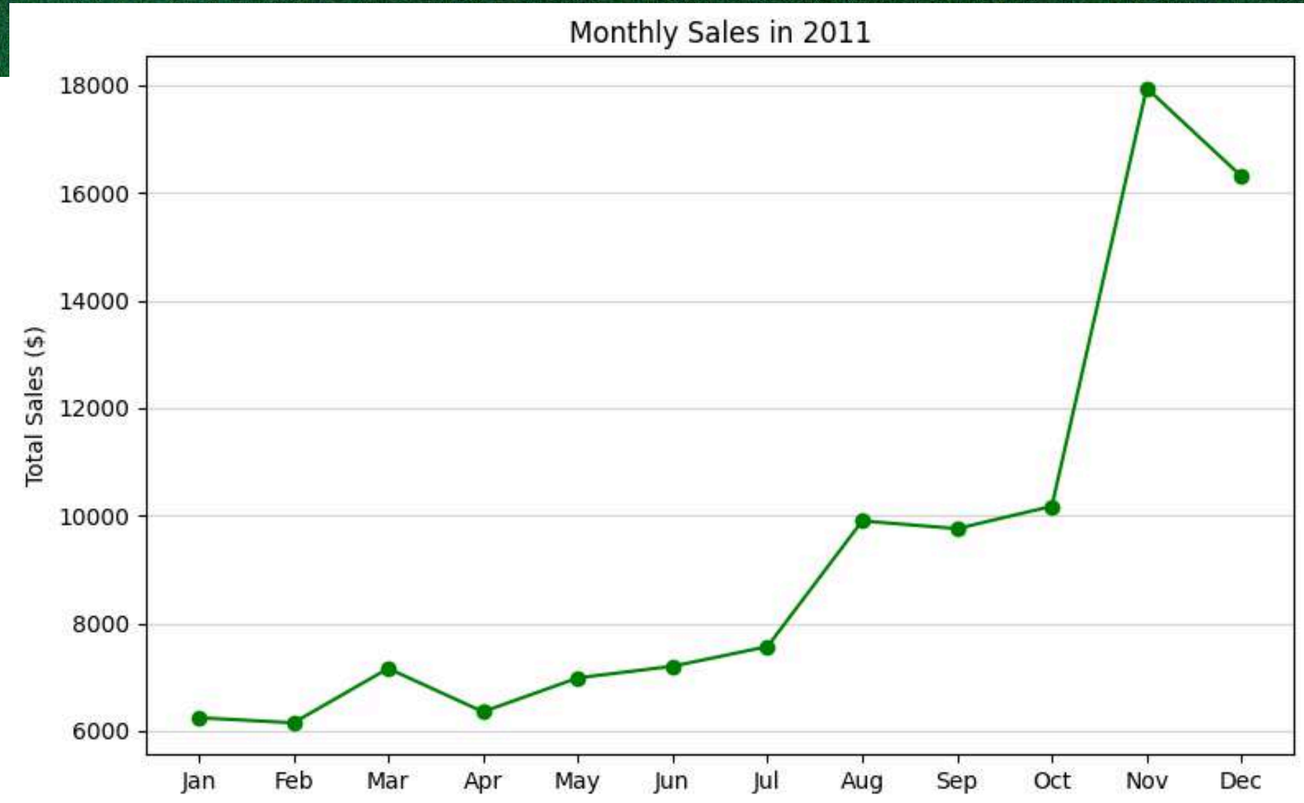
The scatterplots between Unit Price, Quantity, and Sales show us that:

1. There is a positive correlation between Quantity and Sales. **The more items are sold, the higher the number of sales.**
2. There are **no correlation** between **Quantity vs Unit Price** and **Unit Price and Sales**.

Based on patterns from the heatmap graph, it can be concluded:
1. There is **a strong positive correlation** between **Sales and Quantity** with value more than 0.60.
2. There is nearly **no correlation** between **Unit Price vs Quantity** and **Unit Price vs Sales** since their values are less than 0.10.



Spike Revenue Growth from October to November



Generally, in 2011, the company showed **very significant growth.**

From **January to July**, sales were **relatively stable**, but in **August,** transactions skyrocketed from **10,000 USD** to **nearly 18,000 USD** in **November**, and then decreased slightly to almost **16,000 USD** in **December**. Nevertheless, it was still considerably high.

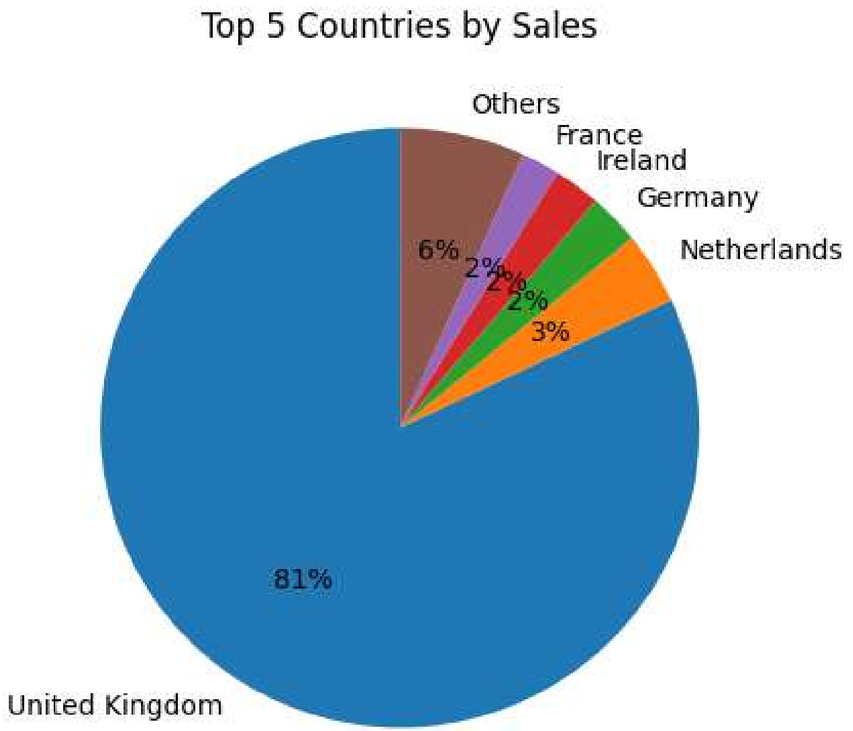
The **increasing numbers in sales** might be caused by **summer holidays** and other events such as **Halloween and Christmas**, where giving presents is the norm.

Top 5 Countries by Sales with UK Dominating The Market

The pie chart illustrates the distribution of the **top 5 markets** and the other markets. **UK leads 81% of the company's revenue**, in contrast to the other countries combined.

There are 3 possibilities why the market is still centered in the UK:

1. **The shipping cost is way cheaper** in the UK compared to other nations;
2. **Low awareness of e-commerce's existence** in other nations compared to the UK;
3. **International customers prefer purchasing gifts from physical stores** rather than from an online store where one has to wait for days/weeks just for the items to be delivered.

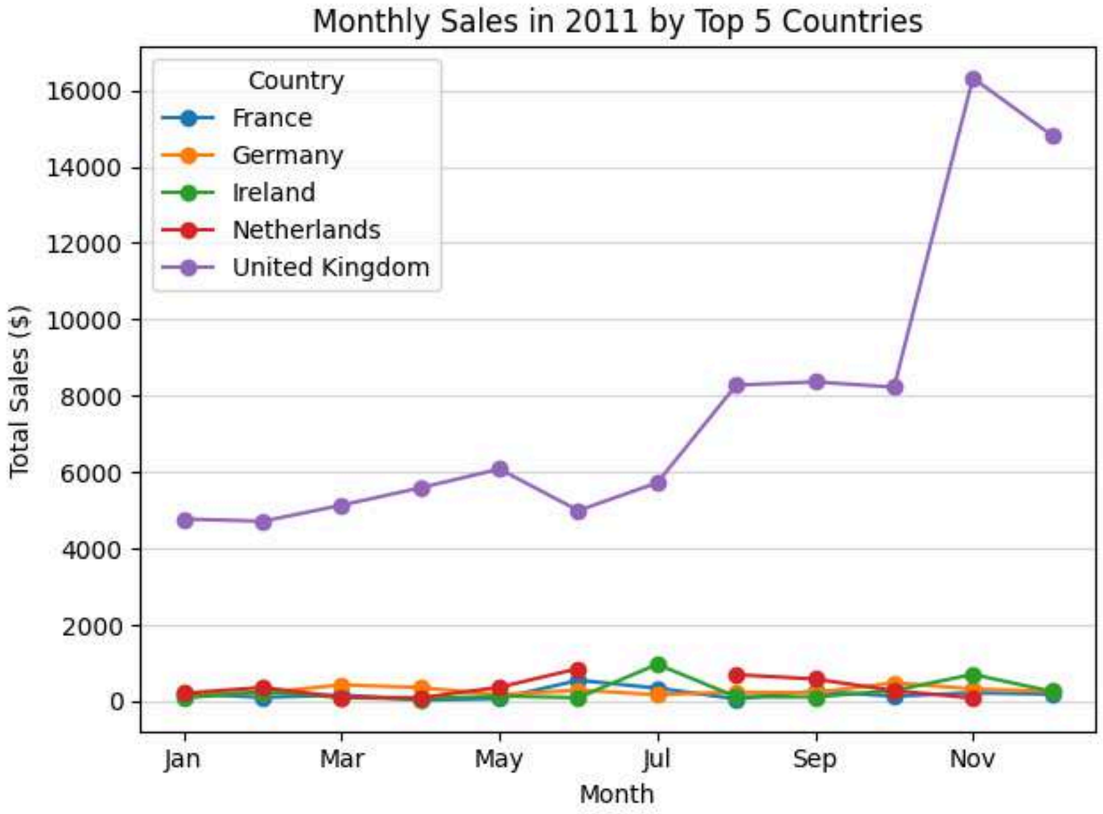


Monthly Sales by Top 5 Countries

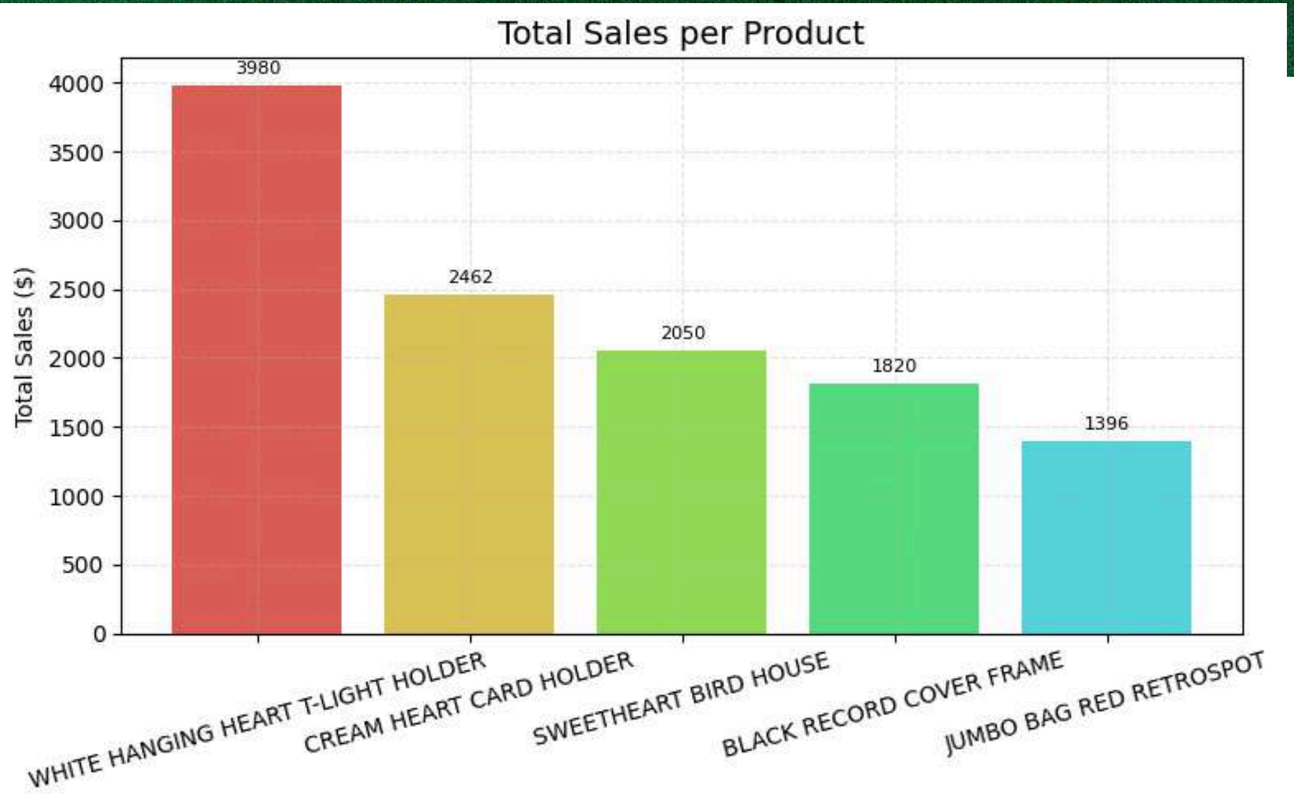
Among the top 5 countries by sales, the **UK shows robust performance**, **peaking in November** and **falling in June**.

The Netherlands has no sales in July or December, in contrast to **Ireland**, which **performs best in July**. On the other hand, **Germany's and France's sales are mostly stable** throughout the months.

In the Netherlands, **people tend to spend time offline** with families, friends, and loved ones during holidays. This is reflected in the government's policy, where **no businesses are open except offline retail** during the summer holiday **in July** and the winter holiday **in December**; therefore, the company made no revenue in those respective months.

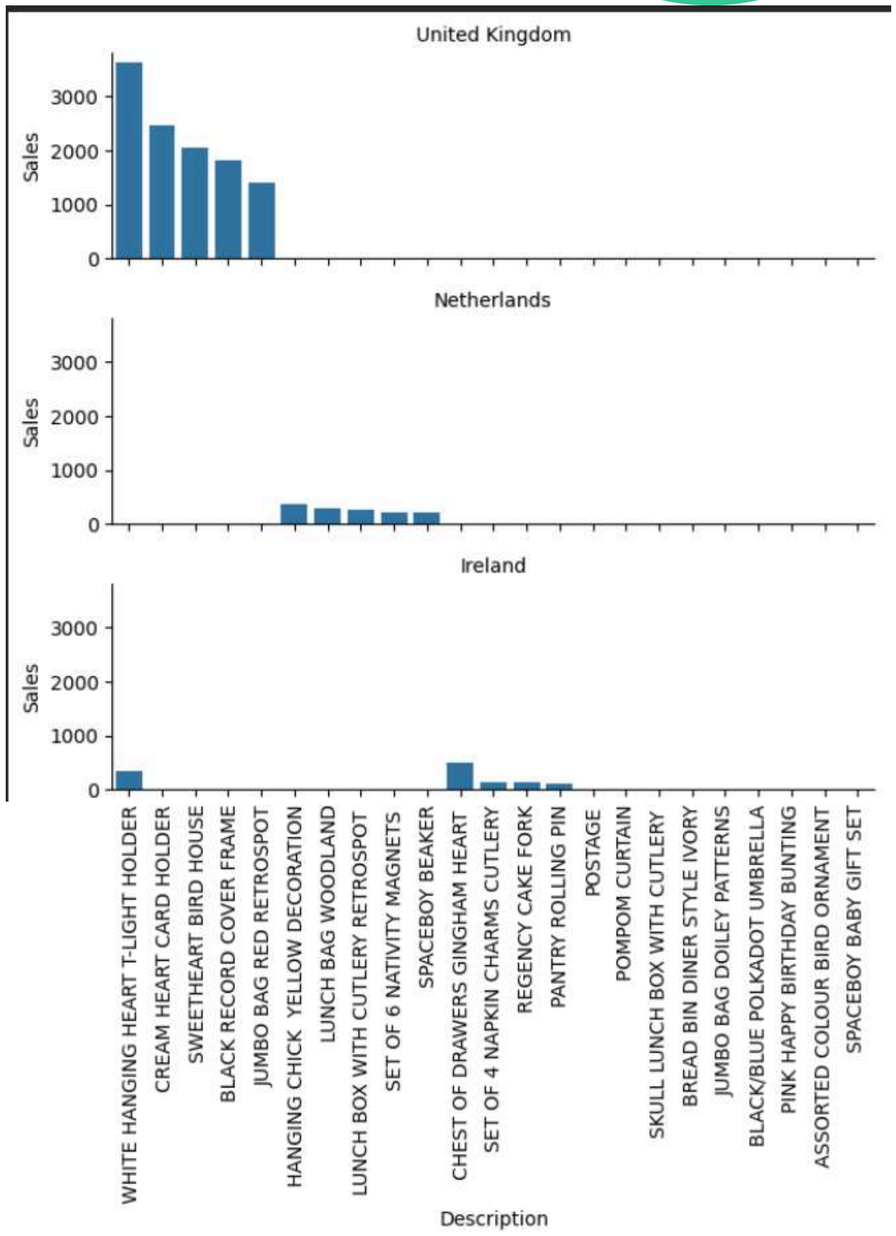


White Hanging Heart T-Light Holder Becomes Best-Selling Products



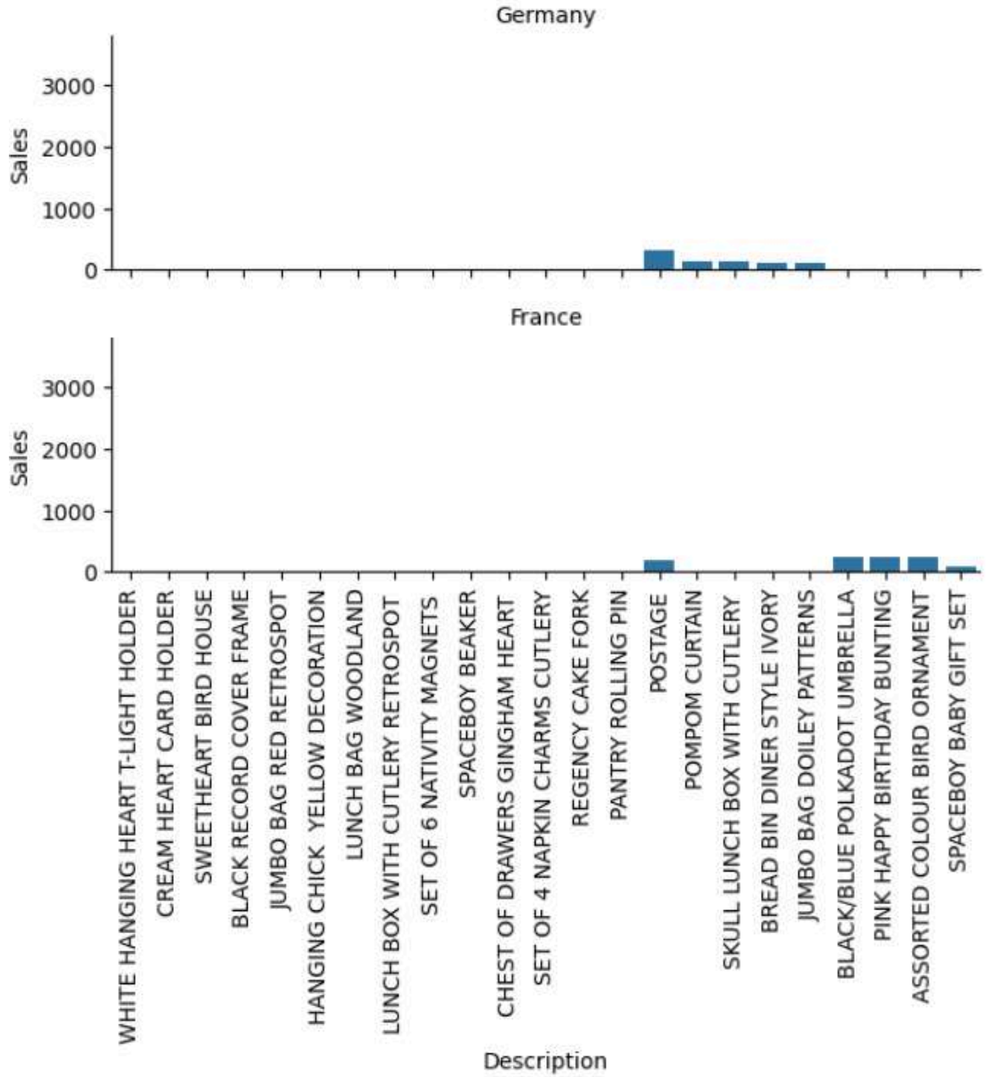
Among the 5 best-selling products, the **White Hanging Heart T-Light Holder** is the most popular. It made **3,980 USD** in sales and **1,291 purchase orders**.

This T-Light Holder is a unique and thoughtful decoration for special occasions such as parties, celebrations, romantic dinners, or simply as a room accessory. Since **the quantity is significantly higher** than that of the other items, there are some indications of **bulk purchases** of T-Light Holder products.



T-Light Holder is Popular Among British and Irish

Each region has its own preferences for popular products, yet the **White Hanging Heart T-Light Holder generated the most sales in the UK and Ireland**. Both countries share a tea culture where tea lights are usually purchased for this kind of occasion.



Postage Product is More Popular in Germany and France

This indicates that postal collections and snail mail are still popular among people in both countries.

Business Recommendations

1. Increase product stock and warehouse capacity support during peak seasons (e.g., summer holidays, Halloween, Christmas) to prevent understocking and warehouse overload. Push the marketing campaigns during this time to boost sales.

2. Put the best-selling products on the homepage where web visitors can easily spot them. Customize the recommendations according to regional preferences.

3. Experiment with A/B testing for less popular items by offering voucher discounts and bundling options that contain both less popular products and popular products as the leverage.

4. Offer free shipping options with minimum orders to increase international sales.

5. Adjust the payment options and delivery options according to the region's currency, payment systems, and local logistics services.

6. Examine the patterns that indicate wholesale purchasing and separate them from retail purchasing. Propose a B2B partnership to customers who frequently order in large bulk.

Limitations / Challenges

1. The data is only available in one year (2011). To uncover the patterns more accurately, it needs at least three consecutive years.

2. No customer ID column that indicates whether a transaction is made by a different person or a repeated order by the same person.

3. No unit column to distinguish the measurement unit, which can mislead the sales quantity of popular items over less popular items during the analysis process.

4. The analysis is based on the internal data only. Further information about the cause of the low sales trend (or even no sales at all, like what happened in the Netherlands) does not include external factors such as brand awareness, shipping costs, the government's trade policies, website traffic, ad ROI, etc.

Thank You

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Shalsabila Sofiana Nur

alrenasalsa

sofia.shalsabila@gmail.com